

Notes: Acharya, Eisert, Eufinger, and Hirsch (RFS, 2019)
Whatever It Takes: The Real Effects of Unconventional Monetary Policy

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1 Big picture

1.1 Question

The paper asks whether the European Central Bank’s 2012 Outright Monetary Transactions (OMT) announcement restored bank credit supply in a way that translated into real economic activity. The answer is nuanced. OMT improved the market value of European banks through capital gains on their sovereign bond portfolios, but some weak banks used the revived lending capacity to evergreen distressed borrowers rather than to finance productive firms.

The central question is therefore not simply whether unconventional monetary policy raised bank lending. It is whether the credit was allocated to firms that could use it productively.

1.2 Setting

The setting is the eurozone sovereign debt crisis. In July 2012, Mario Draghi announced that the ECB was ready to do “whatever it takes” to preserve the euro. The OMT program allowed the ECB to purchase distressed member states’ sovereign bonds in secondary markets, conditional on a program. Although OMT was never activated, the announcement itself sharply reduced yields on periphery sovereign bonds and increased their prices.

This created a stealth recapitalization of banks that held large amounts of periphery sovereign debt. Banks with large GIIPS sovereign bond holdings realized capital gains when those bonds appreciated. The shock therefore worked through bank balance sheets before reaching firms.

1.3 Core mechanism

The paper studies a three-step mechanism:

1. **Asset-price channel:** OMT raises prices of stressed sovereign bonds.
2. **Bank-capital channel:** banks holding those bonds receive windfall gains and regain lending capacity.
3. **Credit-allocation channel:** weakly capitalized banks direct new lending toward distressed incumbent borrowers, creating zombie lending and credit misallocation.

The crucial distinction is between a healthy bank-lending channel and an unhealthy zombie-lending channel. The same policy can increase credit supply and still fail to raise employment or investment if the marginal credit goes to firms that use it to survive rather than to expand.

1.4 Main results

The paper finds the following results.

- GIIPS banks benefited much more from OMT than non-GIIPS banks. Their equity increased by about 8% from the OMT-induced sovereign bond gains, compared with about 1% for non-GIIPS banks.

- Banks with larger OMT windfall gains increased lending after the announcement, but mainly to existing borrowers rather than to new borrowers.
- The lending increase was concentrated among low-interest-coverage firms, not among high-quality borrowers.
- Weakly capitalized high-gain banks extended subsidized credit to distressed existing borrowers. This is the paper’s core evidence of zombie lending.
- Firms receiving OMT-induced credit generally increased cash and debt, but did not significantly increase employment, investment, or return on assets.
- Zombie firms survived temporarily, but their default rates rose sharply after 2014. Zombie-lending banks also experienced larger increases in nonperforming loans.
- Nonzombie firms in industries with a high zombie prevalence suffered higher interest payments, lower investment, and lower employment growth.

1.5 Economic takeaway

Unconventional monetary policy can stabilize financial markets without producing a strong real recovery if the banking sector remains undercapitalized. OMT reduced sovereign spreads and recapitalized banks indirectly, but the policy did not fully repair weak banks. Those banks then had incentives to use new lending capacity to keep distressed borrowers alive, which distorted competition and crowded out productive firms. The key lesson is that a stealth recapitalization is not a substitute for adequate bank recapitalization and resolution.

2 Incremental contribution of the paper

2.1 Relative to the literature on unconventional monetary policy

A large literature studies whether unconventional monetary policy increases bank lending. This paper asks a second-order but economically central question: *who receives the marginal credit?* The result is that bank lending can increase while real effects remain weak because the allocation of credit deteriorates.

The paper therefore moves beyond the quantity of lending and studies the quality and allocation of lending. It shows that the same balance-sheet relief that expands lending can also revive risk-shifting incentives for banks that remain weakly capitalized.

2.2 Relative to bank bailout and zombie lending papers

Earlier zombie lending studies focus on Japan and on explicit bailouts. The paper extends this logic to a eurozone unconventional monetary policy intervention. OMT was not a direct bank bailout; it was an announcement about sovereign bond purchases. Yet because banks held large amounts of sovereign bonds, it had a bank-recapitalization effect.

The paper’s incremental contribution is to show that zombie lending can arise after an *indirect* bank recapitalization. It also shows why the magnitude of recapitalization matters. If the windfall

gain is large enough to restore strong capitalization, banks lend to healthy borrowers. If it is large enough to restore lending capacity but not large enough to eliminate undercapitalization, weak banks can use the additional capacity to evergreen zombie borrowers.

2.3 Relative to bank lending channel identification

The paper uses a modified Khwaja and Mian bank lending design. By comparing lending by different banks to similar firm clusters, the authors separate credit supply from borrower credit demand. They then connect bank-level treatment intensity to firm-level financial and real outcomes and finally to industry-level spillovers. This creates a complete chain from policy announcement to bank balance sheets, loan allocation, firm behavior, bank losses, and market distortions.

2.4 Why the paper matters

The policy implication is important: restoring financial stability is necessary but not sufficient. If banks remain weak, they may misallocate credit toward incumbent weak borrowers. Monetary policy then stabilizes the banking sector in the short run but can slow restructuring and delay recovery in the real economy.

3 Data and measurement

3.1 Bank-firm loan data

The paper uses Thomson Reuters LPC DealScan to identify bank-firm lending relationships in the European syndicated loan market. The sample covers fiscal years 2009 to 2014. Loans are aggregated to the bank parent, and banks enter the sample if they acted as syndicated loan lead arrangers and were included in the European Banking Authority’s June 2012 capital exercise.

The use of syndicated loans is important because DealScan provides lender identities and loan terms. The downside is that it covers large firms rather than the entire corporate sector. The paper mitigates this by using Amadeus firm data and bank-level balance sheet data to trace the mechanism.

3.2 Bank data

The bank data combine several sources:

- European Banking Authority data on bank sovereign debt holdings before OMT;
- DataStream sovereign bond prices, yields, maturities, and durations;
- Markit sovereign and bank CDS spreads;
- SNL bank balance sheet information.

The EBA sovereign holdings are essential because they allow the authors to measure how exposed each bank was to the OMT-induced sovereign bond price increase.

3.3 Firm data

Firm accounting data come from Bureau van Dijk’s Amadeus database. The paper uses firm-level variables such as assets, leverage, net worth, tangibility, interest coverage, EBITDA/assets, cash holdings, employment, investment, return on assets, and productivity.

The firm data allow the paper to study whether firms receiving post-OMT credit increased real activity or merely adjusted financial balances.

3.4 Key bank treatment: OMT windfall gain

The main bank treatment variable is the OMT windfall gain. Conceptually, it measures how much a bank’s equity increased because OMT raised the value of its sovereign bond portfolio:

$$\text{OMT windfall gain}_b = \frac{\text{Value gain on EU sovereign debt holdings}_b}{\text{Total equity}_b}. \quad (1)$$

The authors calculate bond price changes around the three OMT announcement dates and multiply those price changes by each bank’s sovereign debt holdings by country and maturity. Banks with above-mean OMT windfall gains are called high-gain banks; banks with below-mean gains are called low-gain banks.

3.5 Weakly capitalized banks

The paper classifies a bank as weakly capitalized if its regulatory capital buffer is less than 2 percentage points above the EBA regulatory capital requirement after OMT. The authors also examine robustness to alternative thresholds.

This classification matters because zombie lending incentives are predicted to be strongest when a bank has regained enough capacity to lend but remains weak enough to want to avoid recognizing losses.

3.6 Firm quality: high-IC and low-IC firms

A firm is classified as low quality if its 2009-2011 median interest coverage ratio is below the country median. The interest coverage ratio is:

$$IC = \frac{EBIT}{\text{Interest expense}}. \quad (2)$$

Low-IC firms are more likely to be financially distressed because operating earnings are insufficient or barely sufficient to cover interest obligations.

3.7 Zombie firm classification

The paper classifies a firm as a zombie if it satisfies three conditions:

1. it receives subsidized credit, meaning its interest expense is below a benchmark rate paid by very high-quality firms;
2. its inferred rating is BB or lower;
3. its syndicate composition has either remained constant or banks leaving the syndicate are not replaced by new banks, so the same lending syndicate has an existing exposure.

The third condition is important because zombie lending is an evergreening incentive. Existing lenders have an incentive to keep the borrower alive in order to avoid loss recognition.

3.8 Indirect OMT windfall gain at the firm level

To measure how much a firm benefits indirectly from OMT through its banks, the paper constructs:

$$\text{Indirect OMT windfall gain}_{ijht} = \sum_{l \in L_{ijht}} \frac{\text{Average OMT windfall gain}_{lijh} \times \text{Loan amount}_{lijht}}{\text{Total loan amount}_{ijht}}. \quad (3)$$

This variable is the loan-amount-weighted exposure of a firm to banks that experienced OMT windfall gains. It is used in firm-level outcome regressions.

4 Empirical design and identification

4.1 Step 1: OMT as a bank-balance-sheet shock

The first empirical step documents that banks with larger GIIPS sovereign bond holdings realized larger OMT windfall gains. This part of the design relies on the fact that the announcement unexpectedly increased periphery sovereign bond prices. Banks holding more of those bonds experienced larger capital gains.

This is not a generic eurozone recovery measure. It is a cross-bank treatment because banks differed in their ex ante sovereign bond holdings.

4.2 Step 2: Bank lending channel

The main lending regression is a modified Khwaja and Mian design. For bank b , firm cluster m , and quarter t , the intensive-margin loan volume regression is:

$$\Delta Volume_{bmt+1} = \beta_1 \text{OMT windfall gain}_b \times PostOMT_t + \gamma X_{bt} + \text{FirmCluster}_m \times \text{QuarterYear}_{t+1} + \text{FirmCluster}_m \times \text{Bank} \quad (4)$$

Firm clusters are based on country, industry, and rating. The cluster-time fixed effects absorb time-varying loan demand for similar firms. The cluster-bank fixed effects absorb stable lender-borrower cluster relationships.

The extensive-margin regression replaces the dependent variable with a new-loan dummy for lending to firm clusters without a pre-OMT relationship.

4.3 Step 3: Borrower quality split

The paper augments the lending design with interactions for low-IC borrowers. The goal is to see whether banks with high OMT windfall gains lend to healthy borrowers or to weak borrowers. The results show that the increase in lending is concentrated among existing low-IC borrowers.

4.4 Step 4: Zombie lending test

The paper then distinguishes zombie lending from a healthy firm balance-sheet channel. If the OMT announcement improved borrower fundamentals, low-quality firms might receive new loans at fair prices and then increase real activity. If zombie lending dominates, weak banks subsidize distressed borrowers, and those borrowers do not increase real activity.

The zombie lending regression interacts OMT windfall gains, the post-OMT period, weak capitalization, and zombie status. The key term is:

$$\text{OMT windfall gain}_b \times \text{PostOMT}_t \times \text{Weakly capitalized}_b \times \text{Zombie}_{mt}. \quad (5)$$

A positive coefficient means that banks which both benefited from OMT and remained weakly capitalized increased lending to zombie borrowers.

4.5 Step 5: Firm outcomes

Firm-level outcomes are estimated using:

$$y_{ijht+1} = \beta_1 \text{Indirect OMT windfall gain}_{ijh} \times \text{PostOMT}_t + \gamma X_{ijht} + \text{Firm}_{ijh} + \text{Industry}_h \times \text{Country}_j \times \text{Year}_{t+1} + u_{ijht+1}. \quad (6)$$

The firm fixed effects absorb time-invariant firm heterogeneity. Industry-country-year fixed effects absorb macroeconomic shocks, sector shocks, and national policy changes that affect firms in a specific industry and country.

4.6 Step 6: Externalities from zombie prevalence

To study spillovers, the paper estimates whether nonzombie firms perform worse when their industry-country cell has a higher zombie share:

$$y_{ijht+1} = \beta_1 \text{Nonzombie}_{ijht} + \beta_2 \text{Nonzombie}_{ijht} \times \text{IndustryFracZombie}_{jht} + \gamma X_{ijht} + \text{Firm}_{ijh} + \text{Industry}_h \times \text{Country}_j \times \text{Year}_{t+1} + u_{ijht+1}. \quad (7)$$

The coefficient β_2 measures whether healthy firms suffer in industries with more zombies.

5 Main results in detail

5.1 Bank capitalization after OMT

GIIPS banks realized much larger windfall gains than non-GIIPS banks. Table 2 shows that the average OMT windfall gain is about 8% of equity for GIIPS banks and about 1.1% for non-GIIPS banks. GIIPS banks also had much larger GIIPS sovereign holdings as a fraction of assets.

Bank CDS spreads declined around OMT announcements, especially for GIIPS banks. The paper interprets this as evidence that OMT improved perceived bank stability.

However, the recapitalization was incomplete. Some high-gain banks remained weakly capitalized after OMT. The key friction is therefore not that OMT had no bank-capital effect; it is that the effect was not large or targeted enough to eliminate the risk-shifting incentives of weak banks.

5.2 Loan supply increased, but only at the intensive margin

High-gain banks increased lending after OMT. The modified Khwaja and Mian regressions show a statistically significant increase in loan volume to existing borrowers. In the more restrictive specifications, a 1-standard-deviation higher OMT windfall gain implies roughly a 1.5 percentage point loan volume increase at the intensive margin.

In contrast, there is no significant increase in lending to new firm clusters. This means the OMT-induced credit expansion mainly went to firms that already had relationships with the banks.

5.3 The loan increase went to low-quality borrowers

When firms are split by interest coverage, the loan expansion is concentrated among low-IC firms. This finding is consistent with both a firm balance-sheet channel and zombie lending, so it does not yet prove misallocation. It does, however, show that the marginal borrowers were weak firms.

5.4 Evidence for zombie lending

The paper argues that the evidence favors zombie lending over a healthy recovery of low-quality borrowers. The reasons are:

- low-quality firms received credit at subsidized rates;
- the zombie lending increase came from weakly capitalized banks with OMT windfall gains;
- well-capitalized high-gain banks increased lending to nonzombie borrowers and reduced zombie lending;
- zombie firms did not increase real activity after receiving credit;
- zombie firms later defaulted at higher rates.

The strongest regression result is in Table 5: the quadruple interaction between OMT windfall gain, PostOMT, weak capitalization, and zombie status is positive and significant. This is exactly the pattern expected if weak banks use regained lending capacity to keep impaired borrowers alive.

5.5 Not primarily moral suasion or domestic credit backlog

The authors examine whether governments pressured banks to keep weak domestic firms alive. They find that zombie lending is not concentrated in government-owned banks or government-owned firms. They also find similar patterns for nondomestic borrowers and non-GIIPS banks, although magnitudes are smaller. This evidence points to bank risk-shifting incentives rather than government pressure.

5.6 Firm outcomes after OMT

Firms connected to high-gain banks increased debt and cash holdings. For the full sample, cash and debt increase by similar amounts, suggesting that much of the new credit was retained as cash.

The real variables do not improve. Employment growth, investment, and ROA do not significantly increase. This is central to the paper’s conclusion: OMT restored some bank lending capacity, but the marginal credit did not produce a measurable real expansion.

For zombie firms, debt increases more than cash. The likely interpretation is that zombies use part of the new credit to service other outstanding debt, not to accumulate liquidity or invest.

5.7 Long-run bank and firm losses

Zombie-lending banks experienced larger increases in nonperforming loans after the OMT period. Zombie firms initially defaulted at similar or lower rates than nonzombie firms, consistent with zombie loans keeping them alive, but their default rates rose sharply after 2014.

The mechanism is therefore delay rather than recovery. Zombie lending postpones default recognition, but many zombie firms eventually fail.

5.8 Externalities on healthy firms

Nonzombie firms in high-zombie industries pay higher interest rates, invest less, and have lower employment growth. The effects are economically large. The paper reports that a 7 percentage point increase in the zombie fraction reduces nonzombie investment by about 12.6% of capital and lowers employment growth by about 5.6 percentage points.

The externality is stronger in competitive industries and in industries dependent on external finance. This is consistent with both market-congestion and credit-crowding-out channels.

6 Economic interpretation

6.1 Why weak banks evergreen zombies

A weak bank faces a trade-off. If it recognizes losses on impaired loans, its capital ratio deteriorates, regulatory scrutiny increases, and market confidence can fall. If it instead provides subsidized new credit to an impaired borrower, it can delay default recognition. This strategy is costly in present value but attractive to weakly capitalized banks because it preserves accounting and regulatory capital in the short run.

The OMT announcement gives such banks lending capacity, but not enough capital to remove the underlying incentive problem. That is why zombie lending is strongest for banks that receive high OMT gains but remain weakly capitalized.

6.2 Why credit expansion need not increase real activity

A loan increases real activity only if it finances positive-NPV investment or relieves binding financial constraints on productive firms. If it instead finances interest payments, rollover of existing debt, or cash buffers, the real effect is weak. The paper’s firm-level results show that the increase in debt does not translate into higher investment, employment, or ROA.

6.3 Why zombies hurt healthy firms

Zombie firms distort the market in two ways. First, they absorb bank credit that could have gone to productive firms. Second, they remain active in product and labor markets, depressing product prices and keeping workers and capital tied to inefficient uses. Healthy firms then operate in a more congested industry and may reduce investment and employment.

6.4 Policy lesson

The paper does not argue that OMT was bad. OMT probably prevented a much worse financial crisis. The policy lesson is that financial stabilization and real recovery require a sufficiently capitalized banking sector. A broad indirect recapitalization can stabilize asset prices, but it may need to be combined with targeted recapitalization, supervisory action, loss recognition, or bank resolution to prevent credit misallocation.

7 Extracted figures and tables from the paper

7.1 Table 1: Variable definitions and Table 2: Descriptive statistics of banks around OMT

Read.

- Table 1 defines the treatment variables, bank variables, firm variables, and controls used throughout the paper.
- The key bank treatment is the OMT windfall gain; the key bank weakness measure is weak capitalization; the key firm treatment is indirect exposure to banks with OMT gains.
- Table 2 shows that GIIPS banks had much larger OMT windfall gains and larger GIIPS sovereign exposures than non-GIIPS banks.
- Weakly capitalized high-gain banks remained highly leveraged even after OMT.

Economic message. The descriptive evidence already shows the core asymmetry: OMT helped banks with GIIPS sovereign exposure, but not all treated banks became healthy. The treatment is therefore useful for identifying both credit expansion and the consequences of incomplete recapitalization.

Table 1
Variable definitions

Variable	Definition
Bank-level key explanatory variables	
OMT windfall gain	$\Delta \text{Value EU sovereign debt} / \text{Total equity}$
Weakly capitalized	Dummy equal to 1 if a bank is within 2% of EBA's regulatory capital requirement
High-gain	Dummy equal to 1 if a bank has an above mean OMT windfall gain
PostOMT	Dummy equal to 1 if observation falls into the post-OMT period, which starts end of fiscal year 2012
Zombie	Dummy equal to 1 if (1) a firm received subsidized credit, (2) a firm's rating is BB or lower, and (3) its syndicate has already provided a loan
Bank-level dependent variables	
$\Delta V \text{ volume}$	$\ln(V \text{ volume}_{t+1}) - \ln(V \text{ volume}_t)$
New loan	Dummy equal to one if the bank issued a new loan to a firm cluster to which no relation existed pre-OMT
Firm-level key explanatory variables	
Average OMT windfall gain	Average OMT windfall gain of all lead arrangers in a given syndicate
Indirect OMT windfall gain	Proxy for the extent to which a firm benefited from the OMT announcement through their banking relationships
Industry frac zombie	Asset-weighted fraction of zombie firms in a given industry and country in a given year
Firm-level dependent variables	
ΔDebt	$(\text{Total liabilities}_{t+1} - \text{Total liabilities}_t) / \text{Total assets}_t$
ΔCash	$(\text{Cash}_{t+1} - \text{Cash}_t) / \text{Total assets}_t$
Employment growth (Emp. growth)	$\ln(\text{Employment}_{t+1}) - \ln(\text{Employment}_t)$
CAPX	$(\text{Fixed assets}_{t+1} - \text{Fixed assets}_t + \text{Depreciation}) / \text{Fixed assets}_t$, set to 0 if negative
Return on assets (ROA)	$\text{Net income} / \text{Total assets}$
Productivity	$\log(\text{Sales}) - 2/3 * \log(\text{Employment}) - 1/3 * \log(\text{Fixed assets})$
Interest payment (Int. payment)	Interest expenses
Firm-level control variables	
$\ln(\text{Assets})$	Natural logarithm of total assets
Leverage	$(\text{Total assets} - \text{Total equity}) / \text{Total assets}$
Net worth	$(\text{Total shareholder funds} + \text{Liabilities} - \text{Current} + \text{Noncurrent liabilities} - \text{Cash}) / \text{Total assets}$
Tangibility	$\text{Fixed assets} / \text{Total assets}$
Interest coverage ratio (IC)	$\text{EBIT} / \text{Interest expense}$
EBITDA/Assets	$\text{EBITDA} / \text{Total assets}$

(a) Table 1: Variable definitions

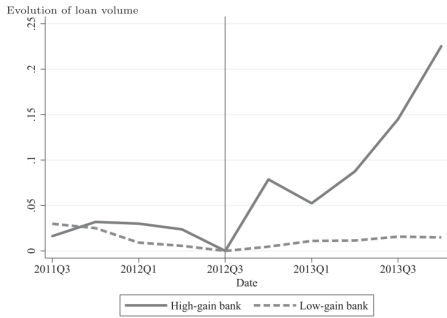


Figure 1

Evolution of loan volume

This figure shows the log-ratio of total loans in a given quarter relative to the quarter of the OMT announcement; that is, the y-axis is normalized to 0 at the time of the OMT announcement, controlling for firm-cluster fixed effects. For each quarter, we aggregate all loans to firms borrowing from banks that are covered by the EBA's June 2012 capital exercise. A bank is classified as high-gain (low-gain) bank if its equity capital increase due to the OMT announcement is above (below) the sample mean. The vertical line represents the OMT announcement period in 2012Q3.

(a) Figure 1: Evolution of loan volume

Table 2
Descriptive statistics of banks around OMT

A. Banks' reaction			
	(1) OMT windfall gain	(2) GIIPS/assets	(3) CDS return
Non-GIIPS banks	0.011	0.010	-0.23 (-9.2)
GIIPS banks	0.08	0.118	-0.96 (-3.4)
t-test for difference	5.69	12.7	7.8
B. Total assets/total equity ratio			
	Precrisis	Crisis/pre-OMT	Post-OMT
Weakly capitalized high-gain	15.3	24.3	19.6
Well-capitalized high-gain	15.0	16.1	13.9
Low-gain	20.8	17.1	16.7
U.S. banks	12.7	9.3	8.7
C. Quasi-leverage ratio			
	Precrisis	Crisis/pre-OMT	Post-OMT
Weakly capitalized high-gain	22.3	71.1	40.4
Well-capitalized high-gain	19.2	26.2	17.9
Low-gain	14.4	31.8	28.6
U.S. banks	8.5	10.1	9.9

Panel A presents descriptive statistics about banks' OMT windfall gain, their GIIPS sovereign debt holdings, and their CDS spread reaction to the OMT announcements. Banks included in the analysis are part of the EBA capital exercise prior to the OMT announcement (June 2012) and must be active in the syndicated loan market during the sample period. GIIPS banks include banks incorporated in Italy, Portugal, and Spain. OMT windfall gain is the value gain on banks' sovereign debt holdings as a fraction of total equity. GIIPS/Assets is the banks' GIIPS sovereign debt holdings as a fraction of total assets. Panel B presents the book leverage ratio for different groups of banks, and panel C presents the quasi-leverage, which is defined as market value of equity plus the book value of debt divided by the market value of equity. Precrisis is defined as the average assets/equity ratios for the years 2004–2006. Crisis/pre-OMT is defined as the assets/equity ratio in the year before the OMT announcement, whereas post-OMT is defined as the assets/equity ratio in the year after the OMT announcement. A bank is classified as high-gain (low-gain) bank if its equity capital increase due to the OMT announcement is above (below) the sample mean. A bank is classified as weakly capitalized (well capitalized) if its leverage ratio is less (more) than 2% above the regulatory threshold post-OMT. t-statistics are reported in parentheses.

(b) Table 2: Descriptive statistics of banks around OMT

Table 3
Loan volume regressions

A. Intensive margin: All firms						
	(1) All banks ΔLoans	(2) All banks ΔLoans	(3) All banks ΔLoans	(4) All banks ΔLoans	(5) All banks Loan inc.	(6) GIIPS banks ΔLoans
OMT windfall gain*PostOMT	0.287*** (4.03)	0.241*** (2.76)	0.253** (2.57)		0.227** (2.17)	0.276** (2.16)
R ²	.098	.736	.774		.702	.763
N	15,930	15,930	15,930		15,930	6,518
B. Extensive margin: All firms						
	New loan	New loan	New loan			New loan
OMT windfall gain*PostOMT	0.086 (1.12)	0.020 (0.65)	0.022 (0.74)			-0.009 (-0.19)
R ²	.048	.594	.627			.669
N	18,291	18,291	18,291			8,655
Bank-level controls	Yes	Yes	Yes		Yes	Yes
Firm-cluster fixed effects	Yes	No	No		No	No
Time fixed effects	Yes	No	No		No	No
Bank fixed effects	Yes	Yes	No		No	No
Firm-cluster-bank fixed effects	No	No	Yes		Yes	Yes
Firm-cluster-time fixed effects	No	Yes	Yes		Yes	Yes
C. Intensive margin: Quality split						
	All banks ΔLoans	All banks ΔLoans	All banks ΔLoans	All banks ΔLoans	All banks Loan inc.	GIIPS banks ΔLoans
OMT windfall gain*PostOMT	0.039 (0.56)	-0.064 (-1.01)	-0.045 (-1.04)			
OMT windfall gain*PostOMT *Low-IC	0.348** (2.50)	0.286** (2.07)	0.251** (2.12)	0.309** (2.30)	0.319** (2.25)	0.399*** (3.16)
R ²	.098	.737	.778	.772	.706	.778
N	15,930	15,930	15,930	15,930	15,930	6,518

(b) Table 3: Loan volume regressions, first part

7.2 Figure 1: Evolution of loan volume and Table 3: Loan volume regressions

Read.

- Figure 1 plots loan volume relative to the OMT quarter. High-gain banks expand lending after OMT; low-gain banks do not.
- Table 3 begins the formal lending-channel evidence. Panel A shows a positive intensive-margin lending response to the OMT windfall gain.
- The extensive-margin tests show no comparable increase in new lending relationships.
- The quality split indicates that the lending increase is concentrated among low-IC borrowers.

Economic message. The first-stage credit supply effect is real: banks that benefited from OMT lent more. But the increase mainly went to existing relationships and low-quality firms, which already hints that the policy transmission may be distorted.

7.3 Table 3: Loan volume regressions continued and Figure 2: Benchmark interest rates and fraction of zombie firms

Table 3
Continued

D. Extensive margin: Quality split

	New loan	New loan	New loan	New loan	New loan
OMT windfall gain*PostOMT	0.068 (0.77)	0.059 (0.80)	0.053 (0.68)		
OMT windfall gain*PostOMT *Low-IC	0.040 (0.53)	-0.058 (-0.56)	-0.053 (-0.48)	-0.034 (-0.30)	-0.023 (-1.08)
R^2	.048	.595	.627	.633	.676
N	18,291	18,291	18,291	18,291	8,655
Bank-level controls	Yes	Yes	Yes	No	No
Firm-cluster fixed effects	Yes	No	No	No	No
Time fixed effects	Yes	No	No	No	No
Bank fixed effects	Yes	Yes	No	No	No
Firm-cluster-bank fixed effects	No	No	Yes	Yes	No
Firm-cluster-time fixed effects	No	Yes	Yes	Yes	Yes
Bank-time fixed effects	No	No	No	Yes	Yes
Mean dependent variable (panels A&C)	0.023	0.023	0.023	0.023	0.135
SD dependent variable (panels A&C)	0.108	0.108	0.108	0.108	0.329
Mean dependent variable (panels B&D)	0.038	0.038	0.038	0.038	0.036
SD dependent variable (panels B&D)	0.151	0.151	0.151	0.151	0.157
Mean OMT windfall gain	0.029	0.029	0.029	0.029	0.080
SD OMT windfall gain	0.060	0.060	0.060	0.060	0.054

This table presents the results of a modified Khwaja and Mian (2008) bank lending channel regression. The unit of observation is a firm cluster-bank-quarter year. In panels A and C, the dependent variable is the change in loan volume of a firm cluster-bank relation in a given quarter. In panels B and D, the dependent variable is a dummy equal to 1 if a new loan is issued to a firm cluster with which no prior relation existed. Firm clusters are formed based on a firm's country of incorporation, industry, and rating. *OMT windfall gain* is the value gain on banks' sovereign debt holdings as a fraction of total equity. *PostOMT* is an indicator variable equal to 1 starting in quarter four of 2012 and 0 before. A firm is classified as high-IC (low-IC) ratio firm if its 2009–2011 median IC ratio is above (below) the country-specific 2009–2011 median IC ratio. Bank-level controls include the logarithm of total assets, equity/assets, impaired loans/equity, and return on assets. Standard errors are clustered at the bank-level. t -statistics are reported in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

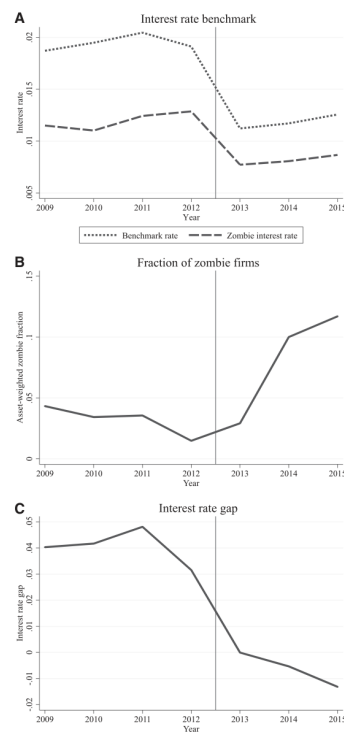


Figure 2

Benchmark interest rates and fraction of zombie firms

Panel A shows the average benchmark interest rate (blue-dotted line) and the median interest rate paid by zombie firms (red-dashed line). Panel B shows the asset-weighted fraction of zombie firms in our sample. Panel C shows the average interest rate gap for firms that only became a zombie post-OMT. We classify a firm as zombie if it meets the following three criteria: (1) the firm received subsidized credit, (2) its rating (derived from 3-year median ICs) is BB or lower, and (3) the syndicate composition has either remained constant, or banks that left the syndicate were not replaced by new participants; that is, the same syndicate has already provided a loan to the firm. The vertical line represents the OMT announcement period in 2012Q3.

(a) Table 3: Loan volume regressions, continued

(b) Figure 2: Benchmark interest rates and fraction of zombie firms

Read.

- The continuation of Table 3 confirms the absence of a significant extensive-margin lending effect, even when low-IC borrowers are considered.
- Figure 2 shows how the benchmark interest rate and zombie interest rate evolve, and how the fraction of zombie firms rises sharply after OMT.
- Panel C of Figure 2 shows that firms becoming zombies post-OMT had paid rates above the benchmark pre-OMT but rates below it afterward.

Economic message. The zombie classification is not simply capturing permanently cheap borrowers. The sharp post-OMT drop in interest-rate gaps for newly classified zombies is consistent with a change in bank lending behavior.

7.4 Table 4: Descriptive statistics and Figure 3: Fraction of zombie loans: High-gain and GIIPS banks

Table 4
Descriptive statistics

A. Breakdown of zombie firms by country

Country	Zombie firms/ Total borrowers	# of firms in sample
Germany	11 (7%)	167
Spain	37 (18%)	201
France	20 (10%)	203
United Kingdom	33 (12%)	286
Italy	40 (20%)	198

B. Difference in group of firms

	High-IC ratio	Low-IC ratio no zombie	Zombie	Difference (2)-(3)
Total assets (mn)	3,130	2,820	2,110	710** (2.05)
Tangibility	0.556	0.598	0.491	0.107*** (6.68)
Int. cov.	6.645	1.790	0.230	1.56*** (8.28)
Net worth	0.309	0.223	0.186	0.037*** (2.91)
EBITDA/Assets	0.134	0.077	0.030	0.047*** (11.92)
Leverage	0.602	0.709	0.742	-0.033*** (-2.84)
Loan amount/Total assets (%)	21.25	18.80	21.92	-3.12 (-1.48)
Maturity (months)	60.03	63.35	61.39	1.96 (0.58)
Term loan(%)	53.31	51.76	54.38	-2.62 (-0.58)

Panel A presents a breakdown of the number of zombie firms by country where the fraction of all sample firms in a given country is shown in parentheses. Panel B presents a test for the difference in means between low-IC nonzombie firms and zombie firms. A firm is classified as high-IC (low-IC) ratio firm if its 2009–2011 median IC ratio is above (below) the country-specific 2009–2011 median IC ratio. For the definition of a zombie firm see the caption of Figure 2. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

(a) Table 4: Descriptive statistics

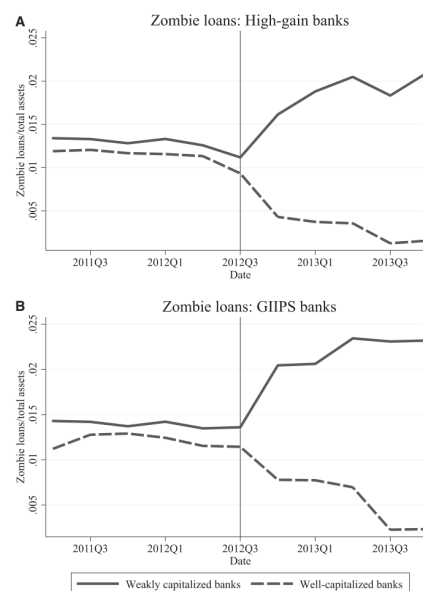


Figure 3

Fraction of zombie loans: High-gain and GIIPS banks

The figure presents the ratio of zombie loans to total assets separately for the subsample of high-gain banks (panel A) and GIIPS banks (panel B). A bank is classified as high-gain (low-gain) bank if its equity capital increase due to the OMT announcement is above (below) the sample mean. A bank is classified as weakly capitalized (well capitalized) if its leverage ratio is less (more) than 2% above the regulatory threshold post-OMT. See the caption of Figure 2 for the definition of a zombie firm. The vertical line represents the OMT announcement period in 2012Q3.

(b) Figure 3: Fraction of zombie loans: High-gain and GIIPS banks

Read.

- Table 4 shows that zombie firms are more leveraged, less profitable, and have much lower interest coverage than other low-IC firms.

- The country breakdown shows that the zombie problem is most severe in Spain and Italy, but it also appears in core countries.
- Figure 3 shows that weakly capitalized high-gain banks and weakly capitalized GIIPS banks sharply increase zombie loans after OMT.

Economic message. The firms classified as zombies look economically impaired, not merely temporarily illiquid. The bank-level figure connects the rise of zombie loans to the precise banks predicted by the theory: high-gain banks that remained weak.

7.5 Table 5: Loan volume regressions - zombie lending and Figure 4: Financial and real effects

Table 5
Loan volume regressions - zombie lending

	(1) All banks Δ Loans	(2) All banks Δ Loans	(3) All banks Δ Loans	(4) All banks Δ Loans	(5) All banks Loan inc.	(6) GIIPS banks Δ Loans
OMT windfall gain*PostOMT ($\hat{\beta}_1$)	0.484*** (3.17)	0.420** (2.30)	0.437** (2.01)			
Zombie*PostOMT	0.003 (1.28)					
OMT windfall gain*Zombie	-0.061 (-1.09)	0.033 (0.07)				
Weakly capitalized*PostOMT ($\hat{\beta}_5$)	0.007 (0.45)	0.008 (0.35)	0.001 (0.48)			
Weakly capitalized*Zombie	0.010** (2.14)	0.009** (1.98)				
Weakly capitalized*PostOMT *Zombie ($\hat{\beta}_6$)	-0.006 (-0.16)	-0.009 (-1.13)	-0.003 (-0.13)	0.008 (0.36)	-0.023 (-1.15)	-0.007 (-0.97)
OMT windfall gain*PostOMT *Zombie ($\hat{\beta}_3$)	-0.562* (-1.93)	-0.511*** (-3.17)	-0.501*** (-3.72)	-0.546*** (-3.96)	-0.553** (-2.32)	-0.684** (-2.59)
OMT windfall gain*PostOMT *Weaklycapitalized ($\hat{\beta}_2$)	-0.494** (-2.48)	-0.445** (-2.10)	-0.438** (-2.04)			
OMT windfall gain*Weakly capitalized*Zombie	-0.042 (-0.56)	-0.062 (-1.37)				
OMT windfall gain*PostOMT *Weaklycapitalized*Zombie ($\hat{\beta}_4$)	0.792** (2.32)	0.733*** (3.55)	0.699** (2.38)	0.661** (2.43)	0.804*** (2.92)	0.980** (2.38)
R^2	.077	.706	.740	.747	.711	.777
N	18,098	18,098	18,098	18,098	18,098	7,815
$\hat{\beta}_1 + \hat{\beta}_3$	-0.078 (-0.98)	-0.091 (-1.14)	-0.064 (-0.85)			
$\hat{\beta}_1 + \hat{\beta}_2$	-0.010 (-0.16)	-0.025 (-0.25)	-0.001 (-0.01)			
$\hat{\beta}_2 + \hat{\beta}_4$	0.298** (1.97)	0.288** (2.08)	0.261* (1.76)			
$\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 + \hat{\beta}_4$	0.220** (2.17)	0.197** (2.29)	0.197* (1.79)			
$\hat{\beta}_5 + \hat{\beta}_6$	0.001 (0.02)	-0.001 (-0.06)	-0.002 (-0.09)			
Mean dependent variable	0.026	0.026	0.026	0.026	0.146	0.031
SD dependent variable	0.121	0.121	0.121	0.121	0.343	0.152
Mean OMT windfall gain	0.029	0.029	0.029	0.029	0.029	0.080
SD OMT windfall gain	0.060	0.060	0.060	0.060	0.060	0.054
Bank-level controls	Yes	Yes	Yes	No	No	No
Time fixed effects	Yes	No	No	No	No	No
Firm-cluster fixed effects	Yes	No	No	No	No	No
Bank fixed effects	Yes	Yes	No	No	No	No
Bank-time fixed effects	No	No	No	Yes	Yes	Yes
Firm-cluster-bank fixed effects	No	No	Yes	Yes	Yes	No
Firm-cluster-time fixed effects	No	Yes	Yes	Yes	Yes	Yes

This table presents the results of a modified Khwaja and Mian (2008) bank lending channel regression used in Table 3. Firm clusters are formed based on a firm's country of incorporation, industry, rating, and whether the firm is a zombie. Hence, clusters consist entirely of zombie or nonzombie firms. *OMT windfall gain* is the value gain on banks' sovereign debt holdings as a fraction of total equity. *Weakly capitalized* is a dummy variable that equals one if a bank is less than 2% above the regulatory capital requirement post-OMT. *PostOMT* is an indicator variable equal to 1 starting in quarter four of 2012 and 0 before. See the caption of Figure 2 for the definition of a zombie firm. Bank-level controls include the logarithm of total assets, equity/assets, impaired loans/equity, and return on assets. Standard errors are clustered at the bank level. *t*-statistics are reported in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

(a) Table 5: Loan volume regressions - zombie lending

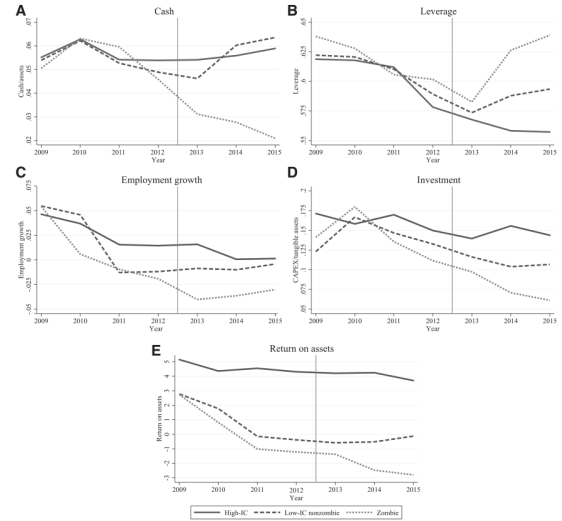


Figure 4
Financial and real effects

Panel A shows the evolution of the asset-weighted cash holdings as a fraction of total assets, and panel B shows the evolution of the asset-weighted leverage as a fraction of total assets. Panel C shows the evolution of the asset-weighted employment growth rates; panel D shows the evolution of the asset-weighted capital expenditures as a fraction of tangible assets; and panel E shows the evolution of the asset-weighted return on assets. All panels show *High Indirect OMT windfall gain* firms (i.e., firms with an above median *Indirect OMT windfall gain*), where we split these borrowers into three groups: High-IC ratio firms (blue-solid line), low-IC ratio nonzombie firms (red-dashed line), and zombie firms (green dotted line). A firm is classified as high-IC (low-IC) ratio firm if its 2009–2011 median IC ratio is above (below) the country-specific 2009–2011 median IC ratio. See the caption of Figure 2 for the definition of a zombie firm. The vertical line represents the OMT announcement period in 2012Q3.

(b) Figure 4: Financial and real effects

Read.

- Table 5 is the paper's central zombie-lending table. The quadruple interaction among OMT windfall gain, PostOMT, weak capitalization, and zombie status is positive and significant.

- Well-capitalized high-gain banks increase lending to nonzombie firms and reduce zombie exposure.
- Figure 4 shows that firms connected to OMT-gain banks increase financial balances, but do not show clear improvements in employment, investment, or ROA.

Economic message. The treatment effect depends critically on bank capitalization. OMT gains are productive only when banks are strong enough to allocate credit to healthy firms. When banks remain weak, the additional lending capacity supports zombies.

7.6 Table 6: Financial and real effects and Figure 5: Evolution of nonperforming loans and firm default rates

Table 6
Financial and real effects
A. All firms

	(1) ΔCash	(2) ΔDebt	(3) ΔCash-ΔDebt	(4) Emp. growth	(5) CAPX	(6) ROA
Indirect OMT windfall gain	0.315**	0.316**	-0.001	0.061	-0.335	0.014
*PostOMT	(2.54)	(2.36)	(-0.01)	(0.62)	(-1.07)	(0.04)
R^2	.522	.605		.804	.644	.587
N	5,324	6,599		6,486	6,492	5,544
B. Quality classification 2009-2011						
Indirect OMT windfall gain	0.136	0.108		-0.023	-0.092	0.185
*PostOMT	(0.92)	(0.53)		(-0.17)	(-0.27)	(0.53)
Indirect OMT windfall gain	0.574***	0.579**	-0.005	0.087	-0.381	-0.498
*PostOMT*Low-IC	(2.85)	(2.09)	(-0.00)	(0.56)	(-0.83)	(-1.22)
R^2	.550	.636		.806	.644	.596
N	5,324	6,599		6,486	6,492	5,544
C. Zombie firms						
Indirect OMT windfall gain	0.543**	0.541**	0.002	0.030	-0.194	-0.339
*PostOMT*Low-IC	(2.41)	(2.50)	(0.01)	(0.32)	(-0.71)	(-1.45)
Indirect OMT windfall gain	-0.548**	0.648***	-1.196***	0.065	0.007	-0.255
*PostOMT*Low-IC*Zombie	(-2.02)	(2.86)	(-3.38)	(0.57)	(0.01)	(-0.84)
R^2	.574	.625		.772	.592	.516
N	5,324	6,599		6,486	6,492	5,544
Mean dependent variable	0.010	-0.005		0.014	0.141	0.035
SD dependent variable	0.064	0.065		0.07	0.175	0.091
Mean indirect OMT windfall gain	0.022	0.022		0.022	0.022	0.022
SD indirect OMT windfall gain	0.032	0.032		0.032	0.032	0.032
Firm-level controls	Yes	Yes		Yes	Yes	Yes
Firm fixed effects	Yes	Yes		Yes	Yes	Yes
Industry-country-year fixed effects	Yes	Yes		Yes	Yes	Yes

Table 6 presents firm-level regression results. The dependent variables are the change in cash holdings, change in leverage, employment growth, investments, and ROA, respectively. In panel B, a firm is classified as high-IC (low-IC) ratio firm if its 2009-2011 median IC ratio is above (below) the country-specific 2009-2011 median IC ratio. *Indirect OMT windfall gain* measures the firms' indirect gains on sovereign debt holdings through their lenders, that is, for each firm, we measure the exposure it has to the value increase in the sovereign debt holdings of the banks from which it received loans. *PostOMT* is an indicator variable equal to 1 starting at the end of fiscal year 2012 and 0 before. See the caption of Figure 2 for the definition of a zombie firm. Firm control variables include the logarithm of total assets, leverage, tangibility, IC ratio, EBITDA as a fraction of total assets, and net worth. All firm-level control variables are lagged by one period. Standard errors are adjusted for heteroscedasticity and clustered at the firm level. t -statistics are reported in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

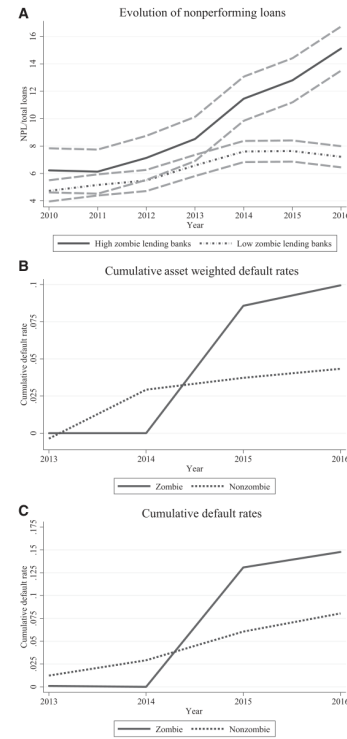


Figure 5

Evolution of nonperforming loans and firm default rates

Panel A shows the asset-weighted fraction of NPLs to total loans (in %) for high and low zombie lending banks by the end of year t . The gray-dashed lines show 95% confidence intervals. A bank is classified as high (low) zombie lending bank if it has an above (below) median fraction of zombie loans/total assets. Panel B presents asset-weighted cumulative firm default rates, and panel C presents equally weighted cumulative firm default rates for zombie (blue-solid line) and nonzombie (red-dashed line) firms by the end of year t . See the caption of Figure 2 for the definition of a zombie firm.

(a) Table 6: Financial and real effects

(b) Figure 5: Evolution of nonperforming loans and firm default rates

Read.

- Table 6 shows that debt and cash rise for firms with higher indirect OMT exposure, but real activity does not rise significantly.

- Zombie firms increase debt more than cash, consistent with using loan inflows to service other debt.
- Figure 5 shows that zombie-lending banks later have higher NPL ratios and zombie firms later experience higher default rates.

Economic message. Zombie lending is not a successful rescue of fundamentally sound firms. It delays default and worsens bank loan performance later. The policy produces short-run survival, not long-run recovery.

7.7 Table 7: Long-run effects and Table 8: Externalities

Table 7
Long-run effects

A. Nonperforming loans

	(1) ΔNPL	(2) ΔNPL	(3) ΔNPL	(4) ΔNPL
OMT windfall gain	0.149 (0.76)	0.128 (0.61)	0.227 (1.26)	0.045 (0.29)
Weakly capitalized	0.017** (2.20)	0.016* (2.05)	0.012 (1.32)	−0.002 (−0.22)
OMT windfall gain*Weakly capitalized	0.980** (2.85)	1.014** (2.75)	0.792** (2.48)	0.714** (2.76)
log(Assets)		−0.001 (−0.90)	0.000 (0.12)	0.002 (0.28)
RWA/TA			0.068* (1.91)	0.153** (2.91)
R^2	0.594	0.595	0.632	0.771
N	49	49	49	49
Country fixed effects	No	No	No	Yes
B. Firm defaults				
	Default	Default	Default	
Low-IC	0.026*** (4.03)	0.024*** (3.60)	0.029*** (3.58)	
Zombie	−0.036*** (−3.11)	−0.034*** (−2.77)	−0.054* (−1.88)	
Zombie*Post	0.059** (2.11)	0.059** (1.98)	0.070** (2.00)	
Low-IC*Post	0.002 (0.43)	0.004 (0.97)	0.000 (0.02)	
R^2	.067	.085	.195	
N	4,993	4,993	4,993	
Industry fixed effects	Yes	No	No	
Country fixed effects	Yes	No	No	
Time fixed effects	Yes	No	No	
Industry-time fixed effects	No	Yes	No	
Country-time fixed effects	No	Yes	No	
Industry-country-time fixed effects	No	No	Yes	

Panel A presents cross-sectional bank regressions. The dependent variable is the change of a bank's NPLs to total loans ratio (average post- minus average pre-2014). *OMT windfall gain* is the value gain on banks' sovereign debt holdings as a fraction of total equity. *Weakly capitalized* is a dummy variable that equals 1 if a bank is less than 2% above the regulatory capital requirement post-OMT. Panel B presents firm panel regressions for the years 2012–2016. *Post* is an indicator variable equal to 1 starting at the end of fiscal year 2015 and 0 before. We classify a firm as defaulted if it is either no longer active or if insolvency proceedings have been opened. A firm is classified as high-IC (low-IC) ratio firm if its 2009–2011 median IC ratio is above (below) the country-specific 2009–2011 median IC ratio. For the definition of a zombie firm see the caption of Figure 2. Standard errors are clustered at the country of bank level in panel A and at the firm level in panel B. *t*-statistics are reported in parentheses. Significance levels: * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

Table 8
Externalities

A. Entire sample

	(1) Int. payment	(2) CAPX	(3) Emp. growth	(4) Productivity
Industry frac zombie*Nonzombie	0.024*** (2.97)	−0.018** (−2.03)	−0.008*** (−2.99)	0.008** (2.32)
R^2	.825	.572	.504	.935
N	5,574	5,070	4,280	4,932
B. Competitive industries				
Industry frac zombie*Nonzombie	0.032*** (3.43)	−0.021** (−2.47)	−0.009*** (−2.92)	0.010** (2.56)
R^2	.836	.589	.572	.957
N	2,800	2,511	2,057	2,421
C. Noncompetitive industries				
Industry frac zombie*Nonzombie	0.025* (1.95)	−0.008 (−1.15)	−0.001 (−0.46)	0.002 (1.07)
R^2	.852	.617	.656	.926
N	2,774	2,559	2,133	2,511
D. External finance-dependent industries				
Industry frac zombie*Nonzombie	0.029*** (3.20)	−0.020*** (−2.84)	−0.009** (−2.11)	0.008** (2.35)
R^2	.821	.583	.521	.920
N	3,032	2,857	2,343	2,618
E. Nonexternal finance-dependent industries				
Industry frac zombie*Nonzombie	0.018* (1.84)	−0.003 (−1.30)	−0.001 (−1.45)	0.002 (1.25)
R^2	.832	.553	.544	.950
N	2,542	2,213	1,937	2,314
Firm-level controls	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes
Industry-country-year fixed effects	Yes	Yes	Yes	Yes

This table presents firm-level regressions. The dependent variables are interest payments, investments, employment growth, and productivity, respectively. *Industry frac zombie* measures the asset-weighted fraction of zombie firms in a given industry and country in a given year (measured using the universe of very large Amadeus firms). *Nonzombie* is an indicator variable equal to 1 for firms not classified as zombie firms. See the caption of Figure 2 for the definition of a zombie firm. To measure the competitiveness of an industry, we use the Herfindahl-Hirschman index (HHI) and split our sample at the median in competitive and noncompetitive industries (panels B and C). To determine an industry's dependence on external finance, we follow Rajan and Zingales (1998) and Acharya and Xu (2017) and consider an industry to be external finance dependent if the median fraction of capital expenditures not financed through internal cash flows of the firms in this industry is above the median of all industries (panels D and E). Firm control variables include the logarithm of total assets, leverage, tangibility, IC ratio, EBITDA as a fraction of total assets, and net worth. Standard errors are adjusted for heteroscedasticity and clustered at the firm level. *t*-statistics are reported in parentheses. * $p < 0.10$; ** $p < 0.05$; *** $p < 0.01$.

(a) Table 7: Long-run effects

(b) Table 8: Externalities

Read.

- Table 7 formalizes the long-run effect: weakly capitalized banks with OMT windfall gains experience larger increases in NPL ratios, and zombie firms have higher later default rates.
- Table 8 shows that nonzombie firms in industries with high zombie prevalence pay higher interest costs, invest less, and have lower employment growth.

- The externality is stronger in competitive industries and in industries more dependent on external finance.

Economic message. The real cost of zombie lending is not limited to the zombies. It spills over to healthy firms through credit misallocation and distorted competition. The outcome is lower aggregate reallocation and weaker recovery.

7.8 Table 9: Effects on nonzombie firms

Table 9
Effects on nonzombie firms

A. Investment loss

Industry	Avg. Investment (% of capital)	Δ Fraction Zombie	investment loss (% of capital)	Investment years lost
Construction	9.48	13.59 pp	24.4	2.6
Manufacturing	11.4	5.50 pp	9.9	0.9
Trade	10.8	9.79 pp	17.6	1.6
Service	11.1	9.28 pp	16.8	1.5
Other	10.6	3.99 pp	7.1	0.7

B. Employment loss

Industry	Avg. emp. growth (%)	Δ Fraction zombie	Employment loss
Construction	-2.1	13.59 pp	10.8 pp
Manufacturing	0.55	5.50 pp	4.4 pp
Trade	0.24	9.79 pp	7.8 pp
Service	-1.0	9.28 pp	7.4 pp
Other	0.5	3.99 pp	3.1 pp

Panels A and B present estimates of the investment and employment losses, respectively, that result from the increased presence of zombie firms in an industry. See the caption of Figure 2 for the definition of a zombie firm. The estimates are derived from a partial equilibrium analysis that compares the outcomes from the de facto fraction of zombie firms to an outcome if the fraction of zombie firms had stayed at its pre-OMT level.

Table 9: Effects on nonzombie firms

Read.

- Table 9 converts the regression estimates into sector-level economic magnitudes for GIIPS countries.
- The construction sector has the largest zombie increase and loses the equivalent of 2.6 years of investment.
- The trade sector loses the equivalent of 1.6 years of investment, and employment losses are also economically large.

Economic message. The estimated externalities are not only statistically significant. They are macroeconomically meaningful. Industries with high zombie prevalence suffer lost investment and employment on a scale that can slow an entire recovery.

8 Identification strengths and limitations

8.1 Strength: policy-induced bank-level heterogeneity

The OMT shock affects banks differently depending on their ex ante sovereign bond holdings. This creates a cross-bank source of variation that is plausibly tied to the announcement rather than to contemporaneous firm demand. The use of pre-OMT sovereign bond holdings is central for identification.

8.2 Strength: loan demand controls

The modified Khwaja and Mian design controls for firm-cluster-time demand. This makes it difficult to explain the lending results purely by low-quality firms demanding more credit. The paper compares different banks lending to similar firm clusters in the same time period.

8.3 Strength: capitalization split

The most convincing evidence comes from the difference between weakly and well-capitalized high-gain banks. If OMT simply improved borrower fundamentals, both groups should lend productively. Instead, weak banks lend to zombies while strong banks lend more to nonzombies.

8.4 Strength: outcome validation

The paper validates the zombie interpretation using downstream outcomes. Zombie firms do not improve real activity and default later at higher rates. Zombie-lending banks later experience higher NPL ratios. These are exactly the predictions of zombie lending.

8.5 Limitations

The sample is built from syndicated loans, so it focuses on large firms and lead arrangers. The results may not fully describe small-firm credit markets. Zombie classification relies on benchmark interest rates and inferred ratings, so misclassification is possible, although the paper provides multiple robustness checks. Finally, OMT occurred in a complex policy environment with LTRO, EBA capital exercises, and national interventions. The paper addresses these concerns, but no empirical design can fully isolate all contemporaneous European crisis policies.

9 Short summary

The OMT announcement stabilized periphery sovereign bond markets and increased the value of bank sovereign bond portfolios. Banks with larger OMT windfall gains increased lending, but mainly to existing low-quality borrowers. When banks remained weakly capitalized after the windfall, they used their renewed lending capacity to provide subsidized credit to zombie firms. These loans kept borrowers alive temporarily but did not increase employment, investment, or productivity. Over time, zombie firms defaulted more and zombie-lending banks accumulated more nonper-

forming loans. Healthy firms in zombie-heavy industries suffered lower investment and employment growth.

10 Conclusion

The paper's central message is that unconventional monetary policy can restore financial stability without generating a robust real recovery if the banking sector remains undercapitalized. OMT worked as a stealth recapitalization by raising sovereign bond prices, but the recapitalization was incomplete for many banks. Weakly capitalized banks then had incentives to evergreen troubled borrowers, producing zombie lending.

The paper therefore changes how one should interpret the bank lending channel. More lending after monetary intervention is not necessarily good news. The real effect depends on whether the banking sector allocates credit to productive firms or to borrowers that should be restructured or exit. In this case, OMT likely prevented a worse crisis, but without stronger recapitalization or resolution measures it also allowed weak banks to misallocate credit.

The broader implication is that central banks and supervisors must coordinate. Monetary policy can stabilize asset prices and bank balance sheets, but if banks remain weak, supervisory forbearance and low rates can sustain zombies. The effectiveness of unconventional monetary policy therefore depends on the health and incentives of the banking sector through which it is transmitted.